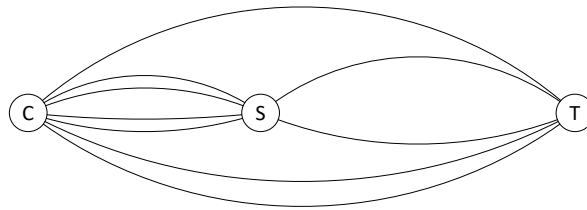


1. Three towns C, S, and T are interconnected as the figure below:



- a) What is the number of different trips* from town C to town T?

*assume that a one-way trip includes any town at most once

- b) What is the number of different round trips from town C to town T and back to town C?

- c) How many trips from part b) include town S at least once? At most once?

2. In IPv4 address consists of 32 bits. They are canonically represented in dot decimal notation which consists of four decimal numbers, each ranging from 0 to 255, separated by dots, e.g., 142.232.55.77. Each address falls into one of the following classes:
- A class – used for the largest networks – begins with 0 which is then followed by a 7-bit network number, and then a 24-bit local address. Network numbers of all 0's or all 1's are not permitted.
 - B class – used for the largest networks – begins with 10 which is then followed by a 14-bit network number, and then a 16-bit local address. Local addresses of all 0's or all 1's are not permitted.
 - C class – used for the smallest sized networks – begins with 110 which is then followed by a 21-bit network number, and then an 8-bit local address. Local addresses of all 0's or all 1's are not permitted.

a) What class is the address 142.232.55.77?

b) How many different addresses of each class are available in IPv4 standard?

3. What value does a variable *counter* have after the following code execution?

```
counter = 100
for i = 1 to 10 {
    counter = counter + 2
    for j = 1 to 20 {
        counter = counter + 3
        for k = 2 to 30 {
            counter = counter + 4
        }
    }
}
```

1

```
counter = 100
for i = 5 to 10 {
    counter = counter + 2
    counter = counter + 3
    for j = 15 to 20 {
        for k = 20 to 30 {
            counter = counter + 4
        }
    }
}
```

2

```
counter = 100
a = 15
b = 20
for i = 1 to 10 {
    a = a + 2
    for j = 1 to 10 {
        b = b + 3
    }
    counter = (i+3)*a + 5*i*b + 4
}
```

3

```
counter = 100
for i = 1 to 10 {
    counter = counter + 2
    for j = 1 to 10 {
        counter = counter + 3
        for k = j+1 to 10 {
            counter = counter + 4
        }
    }
}
```

4

```
counter = 100
for i = 1 to 10 {
    counter = counter + 2
    for j = i+1 to 10 {
        counter = counter + 3
        for k = j+1 to 10 {
            counter = counter + 4
        }
    }
}
```

5

If all of the problems can't be done in the lab period, students can work on them on their own.